

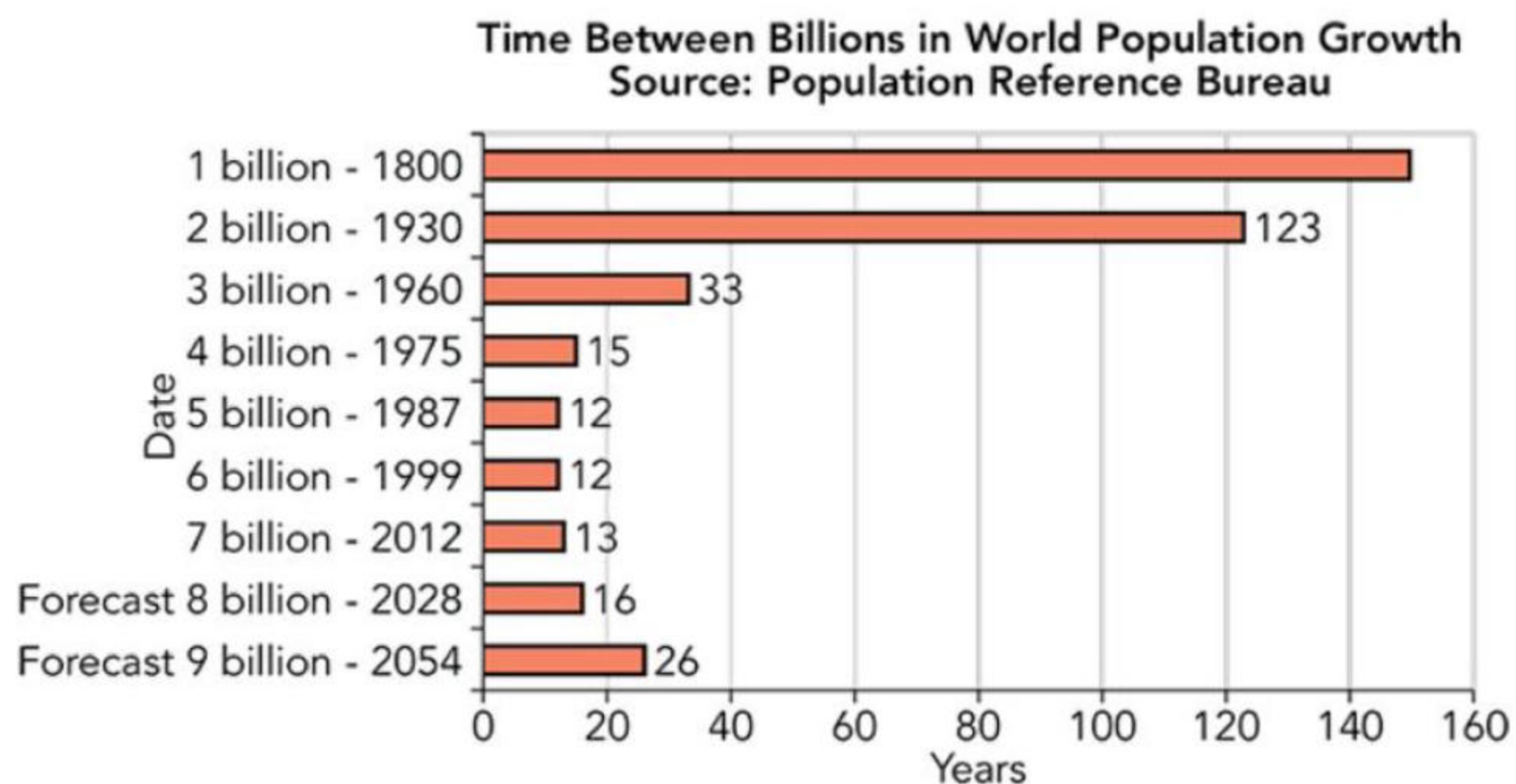
6.1 Population growth and decline

According to the United Nations (UN), the global human population reached 8 billion in November 2022. That is one billion more people in the world than 2011 and two billion more than in 1999. After this period of exponential population growth, **demographers** expect the world population to continue to steadily grow and then stabilise at approximately 9.8 billion people in 2050.

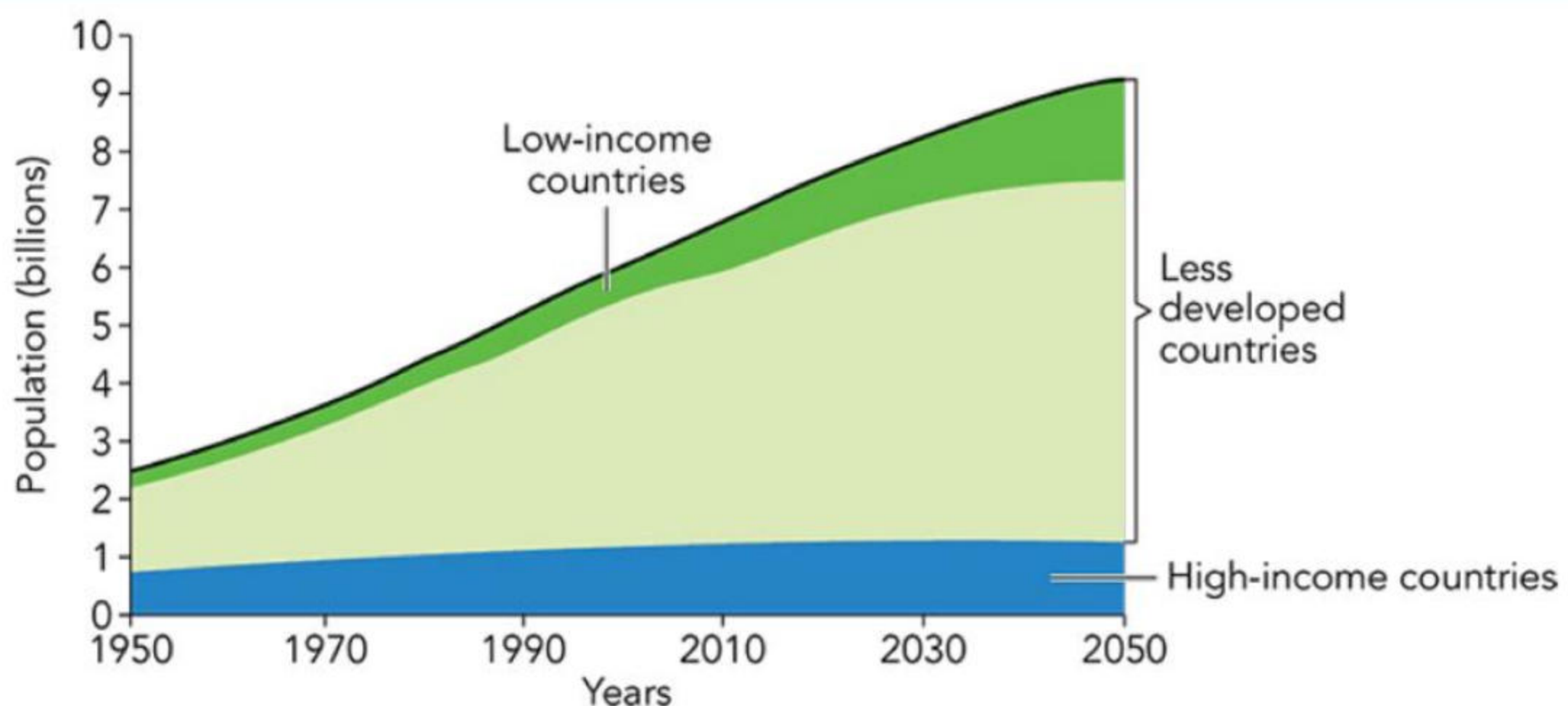
The world reached its highest ever global population growth rate in the early to mid-1960s. Population growth in **Low-income countries (LICs)** reached a high of 2.4% a year. This rapid population growth was called the **population explosion**. By the late 1990s, though, the rate of global population growth declined to 1.8%. By 2023, this had dropped to about 0.88%. However, even though the rate of growth has been falling for several decades, **population momentum** means that the number of people added to the world each year remains very high. In demographic terms, this is because there are many women in the child-bearing age range globally.

The demographic transition, which you will read about later in this topic, took a century to complete in **High-income countries (HICs)**. It has occurred in a single generation (20–30 years) in some LICs. The number of children born has dropped further and faster than most demographers predicted 30 or 40 years ago. Apart from the African continent, where population growth is still nearly 2.4% per year in about 20 countries, birth rates are now declining in virtually every country. According to the Population Reference Bureau:

‘Developed countries as a whole will experience little or no population growth in this century, and much of that growth will be from immigration from less developed countries.’



Source **A**: A graph showing the number of years to add an additional one billion people to the world



Source **B**: Differences in population growth and projected growth in LICs and HICs between 1950 and 2050

- a With the help of Sources A and B, briefly **describe the growth** of the human population over time.
- b It took all of human history until around 1800 years for the world population to reach one billion. **How many years did it take for the world's population to increase from:**
 - i **one to two billion?**
 - ii **six to seven billion?**
- c By 2100, there are likely to be over three billion more people living in the world than today. Suggest two problems which this could cause. Explain why you think each of these problems will be difficult to solve.
- d The figures for 2100 are an estimate. Explain why it is difficult to predict exact population numbers for the future.

TASK 2: Study Source B

- a **Describe the differences** in population growth and projected growth between HICs and LICs between 1950 and 2050.
- b In pairs, discuss possible reasons for why LICs are likely to see the biggest population growth.

‘The milestone is an occasion to celebrate diversity and advancements while considering humanity’s shared responsibility for the planet.’

Source C: Statement by Antonio Guterres, UN Secretary-General, 2022

TASK 3: Study Source C

The UN announced 15 November 2022 as ‘The day of eight billion’ – the date that the world’s population reached eight billion. Do you agree with the statement given by Antonio Guterres, UN Secretary-General?

Write a short speech to justify your response. Then, in small groups, listen to each other’s speeches and try to reach a group conclusion.

The growth and decline in a country’s population

Population change is controlled by three factors: fertility, mortality, and migration. Table 6.1 outlines some of the key measures of fertility and mortality.

Table 6.1: Measures of fertility and mortality

Measure	Definition
Crude birth rate (CBR)	the number of live births per 1000 people per year, usually calculated for a country.
Total fertility rate	the average number of children a woman has during her lifetime.
Crude death rate (CDR)	the number of deaths per 1000 people per year, usually calculated for a country.
Natural increase	the difference between the CBR and CDR. It can be recorded either as a net change per 1000 people or as a percentage.

TIP

It is important to distinguish birth rates from fertility rates. Birth rate is the number of live births per 1000 women in the total population. Fertility rate is the average number of children born to a woman during their reproductive years. The single most important factor in population growth is the total fertility rate.

The population of a country can grow, remain fairly stable, or decline over time. The difference between the crude birth rate and the crude death rate is known as natural increase. If the crude birth rate is higher than the crude death rate, then the population of a country is naturally increasing in size. If the crude death rate is higher than the crude birth rate, then the population of a country is naturally decreasing in size.

The reasons for falling birth rates vary between countries and include advances in women’s rights and freedoms, reduction of infant mortality, and more awareness of contraception. The increasing cost of raising a child has also reduced the fertility rate, particularly in HICs. The reasons for falling death rates in many countries include improved healthcare, disease prevention and treatments, and improved sanitation. **Migration** rates to and from a country also affect the growth and decline of a population.

Migration is the movement of people from one place to another. Migration can take place within a country or internationally and can be a temporary or

permanent move. The balance between the people moving out of a country and the people moving to a country is known as **net migration**. When a country has more people moving to the country than moving away from the country, then the country would have a positive net migration. When a country has more people moving away from the country than moving to the country, then the country would have a negative net migration.

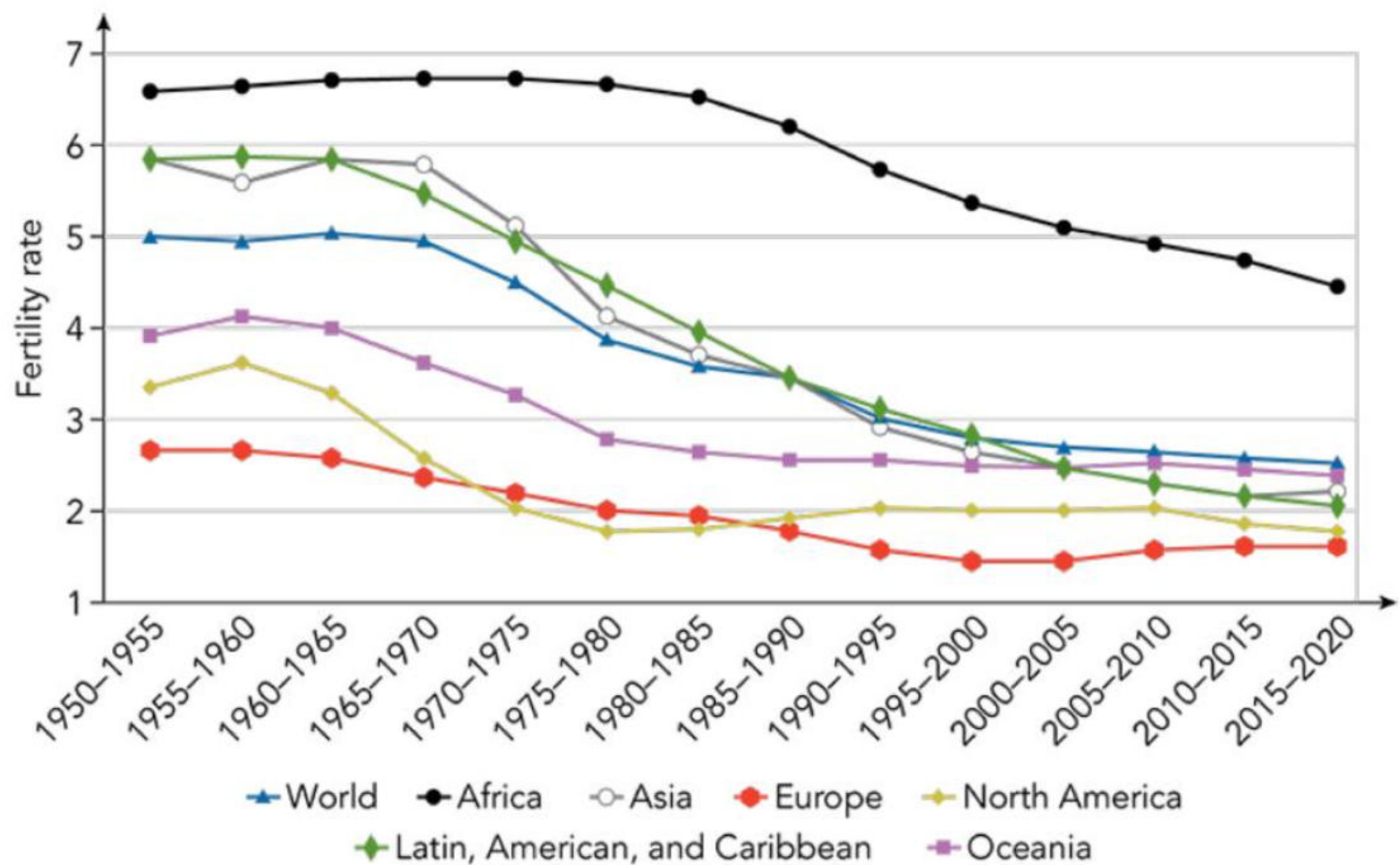


Figure 6.1: Fertility rate in each continent and worldwide, from 1950 to 2020

Table 6.2: Countries with high and low fertility rates. Adapted from World Bank 2021 data.

Country	Total fertility rate
Niger	6.8
Somalia	6.3
Democratic Republic of Congo	6.2
Mali	6.0
Chad	6.3
South Korea	0.8
Malta	1.1
China	1.2
Ukraine	1.2
Spain	1.2

Population policies

A **population policy** is an action taken by a country’s government to influence the size of the population. **Pro-natalist policies** encourage people to have more children. An **anti-natalist policy** aims to slow down a country’s fertility rate by imposing limits on childbirth. These policies can be encouraged by incentives or they can be enforced.

Evaluating a pro-natalist policy

South Korea has pro-natalist policies. Policies include financial incentives, such as more generous housing allowances for families and compulsory paternity leave for fathers. Some of South Korea’s biggest companies are also trying to help by introducing two years of maternity leave and one business is offering a \$75 000 bonus for each baby an employee produces.

However, financial incentives are not working. Between 2006 and 2023, the government spent \$270 billion on trying to increase the fertility rate using cash payments and subsidised childcare services. Despite this, South Korea’s fertility rate fell from 1.1 in 2006 to 0.72 by 2023, far below the replacement rate of 2.1. The problem is that well-off Koreans are also not having children due to pressures to get an elite education, a very good job, and trying to buy a house (all of which are very expensive). You can read about another pro-natalist policy in the DSE about Hungary.

Evaluating an anti-natalist policy

Nigeria has anti-natalist policies. The Nigerian government introduced its national population policy in 1988 (when the fertility rate was 6.23) to try to slow down the rate of population growth and thus improve standard of living. The policy was voluntary and not enforced.

The policy was not very successful as it did not reach rural areas where families often benefit from having many children, and it went against the religious beliefs of some of the population. By 2010, the fertility rate had only declined to 5.61. As a result, the government organised more campaigns to educate people about the benefits of small families, increased the legal minimum age of marriage to 18, and tried to promote the use of contraception. By 2021, the fertility rate had only dropped a little lower to 5.24.

The success of its policies is weakened by the fact that 70% of the Nigerian population is under 30 years old. Girls are often not educated beyond primary school, and family planning is not funded. These factors make it difficult to manage population growth.

TASK 4

In pairs:

- a Discuss how birth rates, death rates, and migration influence rates of population growth.
- b Explain how each of the following can affect population growth rates:
 - i government policy
 - ii net migration
 - iii percentage of women in employment.

TASK 5: Study Sources **D** **E**

- a Explain the difference between a pro-natalist policy and an anti-natalist policy.
- b Study the two different government population policy posters below, Source D and Source E. Which do you think is the poster promoting a pro-natalist policy and which do you think is the poster promoting an anti-natalist policy? Give reasons for your answers.
- c How effective do you think the poster images are at promoting their message?
- d What incentives do you think the population of both countries may need to follow the respective policies?



Sources **D:** Translation: 'One is lonely. The best gift for your child is a younger sibling.'



Sources E: Translation: ‘For a healthy family, wait three years before your second child. You can get these family-planning methods from government health workers, hospitals, and health centres for free.’

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The Demographic Transition Model

The Demographic Transition Model (DTM) shows population change over time, linked to a country’s stage of economic development (Source F).

The original model distinguishes between different demographic phases:

Stage 1: a pre-industrial farming society with no population growth. This is due to a high birth rate and a high death rate. High birth rates occur in societies that rely heavily on agricultural productivity, so they require more people for a larger workforce. High death rates are caused by insufficient medicine and maternal care and inadequate sanitation and public health.

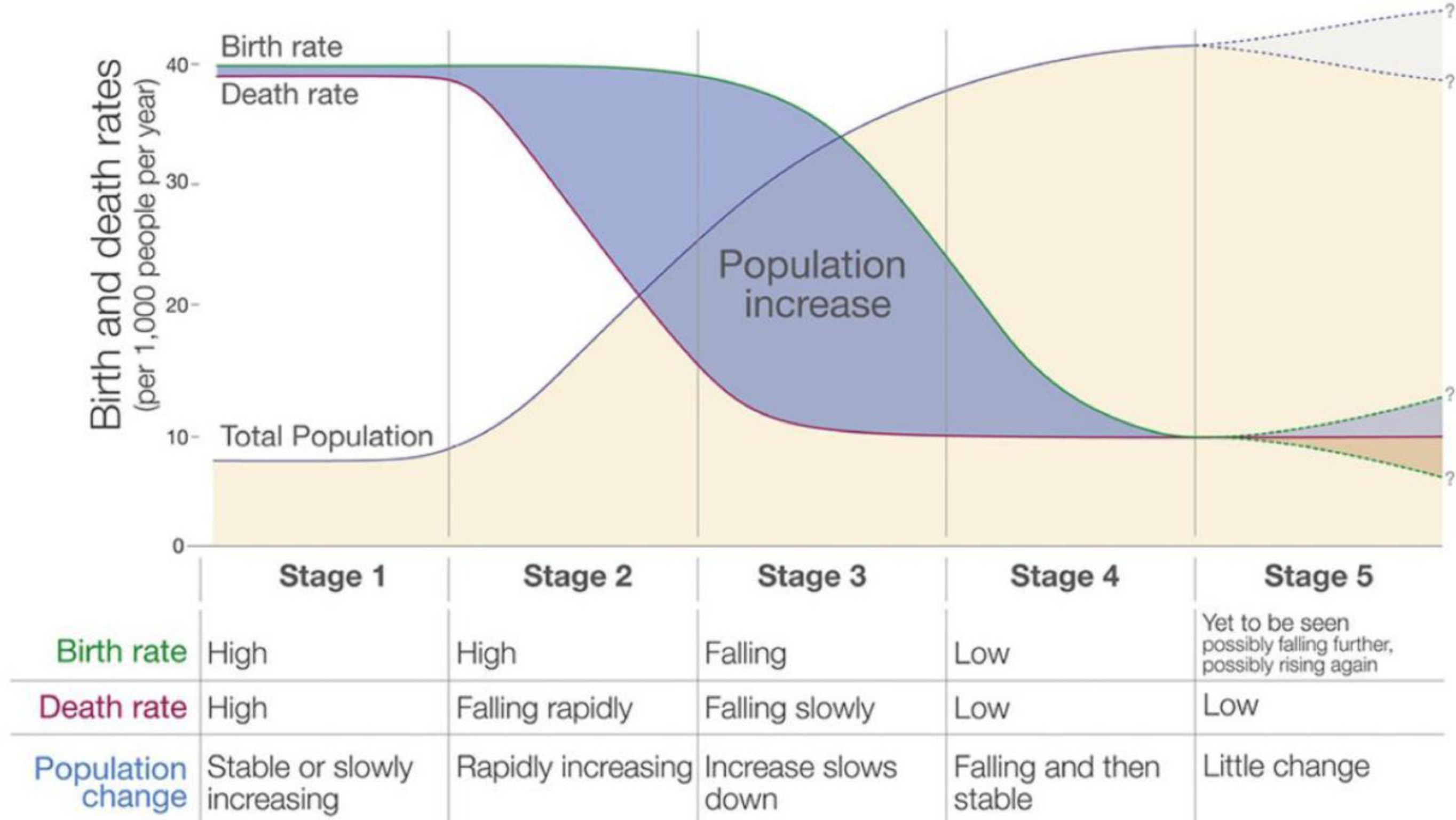
Stage 2: the first phase of industrial development associated with accelerating population growth. The decrease in death rate is mainly caused by improvements in overall health but better education systems, gender equality, and technological advances in food production also decrease the death rate.

Stage 3: a phase of rapid industrialisation. As a result of improved economic conditions, an increase in women’s status and education, and access to contraception, death rates are low and birth rates decrease. Generally the faster the advances in female education and gender equality, the faster the birth rates decline.

Stage 4: a ‘post-industrial’ society, with little or no population growth in the long term. Birth rates and death rates are both low, attributed to strong economies, highly educated citizens, ample healthcare systems, and expanded employment opportunities for women.

Stage 5: birth rates are below death rates and the population size is declining. This Stage was added to the original DTM as, in some HICs, the high costs of raising a family and the increased employment opportunities for women meant that in some HICs, birth rates fall below **replacement level** (2.1 children per woman). This leads to an ageing population and eventually to a decrease in total population.

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Source F: The Demographic Transition Model

As the years pass and a country develops, both the crude birth rate and the crude death rate fall. Population growth slows down, stabilises, and eventually, the population may begin to decrease. The birth rate and death rate fall for a number of reasons:

Reasons for falling birth rates:

- advances in women's rights and freedoms
- reduction of infant mortality
- compulsory education, increasing the age of marriage, and more women in careers
- more awareness and availability of contraception
- the increasing costs of raising a child
- more elderly people have healthcare and pensions, so some people no longer 'rely' on children to care for them in old age.

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Reasons for falling death rates:

- improved food and water quality and quantity
- improved sanitation
- improved healthcare disease prevention and treatments. For example, smallpox has been eradicated through vaccination and improved disease management strategies for HIV means the disease can now be controlled.
- lower **infant mortality rate**
- improved health and safety laws regulating road safety, smoking, and alcohol use.

Strengths and weaknesses of the DTM

The model was based originally on data from England and Scandinavia. Geographical models are useful starting points, but we need to consider them critically.

Strengths

- Model provides an insight into European demographic history. It can be used when speculating about what the future may hold for other countries.
- Model predicts changing birth, death, and population growth rates as countries industrialise. As a model, it can be helpful to understand and predict population trends in countries around the world.

Weaknesses

- The use of numbered ‘stages’ – with European countries all appearing in a ‘top’ class makes the DTM a Eurocentric model (focusing on European history to the exclusion of a wider view of the world).
- The original four-stage model does not consider migration. For example, since the 1970s, Germany’s death rate has exceeded the birth rate. We would therefore expect to see Germany’s population decline after several decades. However, we continue to see an increase in the total population. This population increase is primarily due to immigration into Germany, rather than from people being born inside the country.
- In many countries, the fall in the birth rate in Stage 3 has been at a different rate than the model predicts. For example, in Brazil, it has been less rapid due to religious and political opposition to contraception. In India, it has been more rapid due to its wide-scale female sterilisation programme.
- The model does not consider large-scale global challenges like pandemics or climate change. These events can have significant impacts on birth, death, and migration trends.

TASK 6: Study Source **F**

- Explain three factors that cause fertility rates to be high in some countries but low in others.
- Make a copy of the Demographic Transition Model in Source F. Shade in the natural increase.
- Explain the reasons for declining mortality in Stage 2.
- Suggest why it takes some time before a fall in fertility follows a fall in mortality.
- In pairs, discuss whether you think the strengths of the DTM outweigh the weaknesses of the DTM.